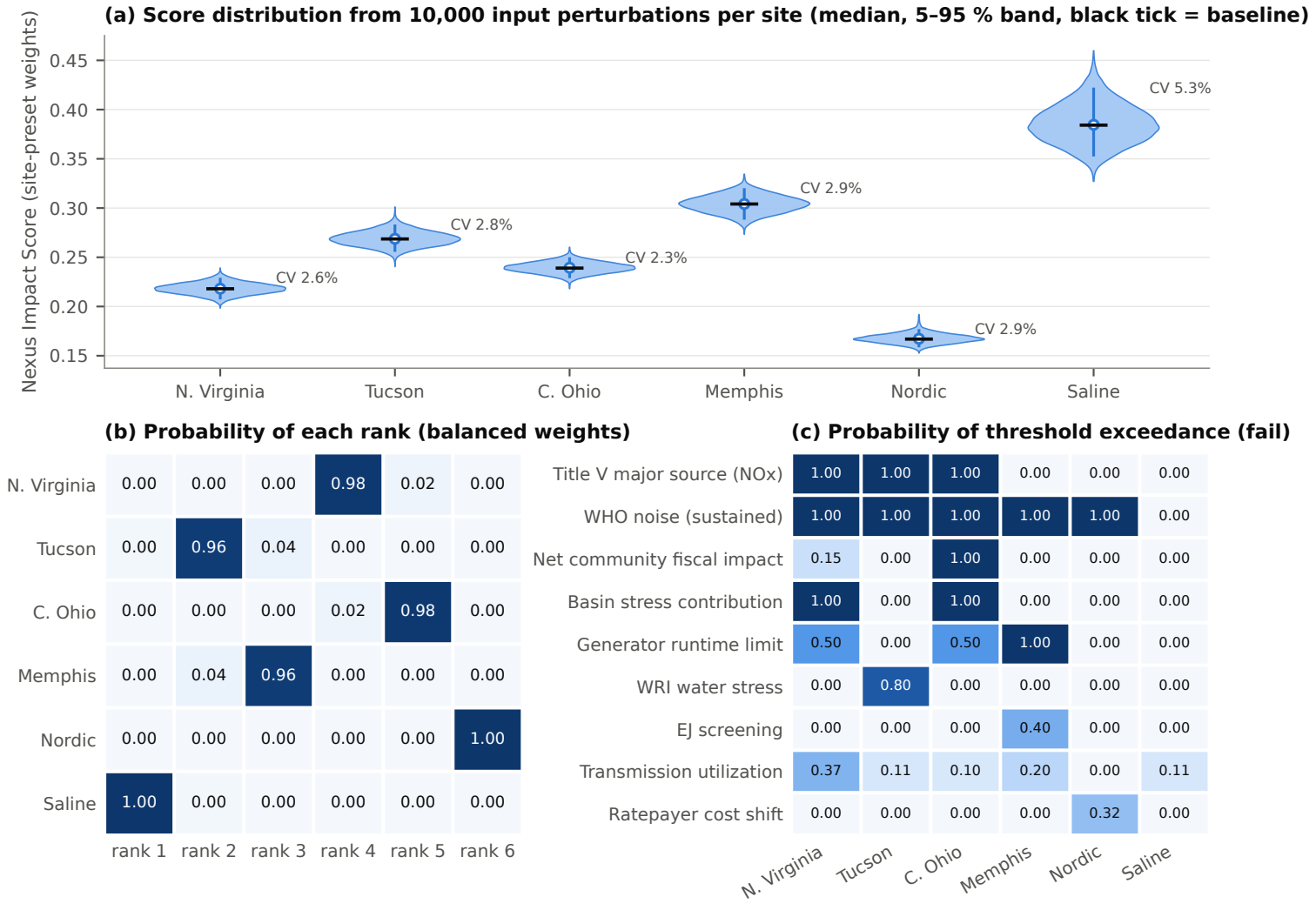

(a) Score distribution from 10,000 input perturbations per site (median, 5–95 % band, black tick = baseline)
Nexus Impact Score (site-preset weights)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
CV 2.6%
CV 2.8%
CV 2.3%
CV 2.9%
CV 2.9%
CV 5.3%
N. Virginia
Tucson
C. Ohio
Memphis
Nordic
Saline
(b) Probability of each rank (balanced weights)
N. Virginia 0.00 0.00 0.00 0.98 0.02 0.00
Tucson 0.00 0.96 0.04 0.00 0.00 0.00
C. Ohio 0.00 0.00 0.00 0.02 0.98 0.00
Memphis 0.00 0.04 0.96 0.00 0.00 0.00
Nordic 0.00 0.00 0.00 0.00 0.00 1.00
Saline 1.00 0.00 0.00 0.00 0.00 0.00
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
(c) Probability of threshold exceedance (fail)
Title V major source (NOx) 1.00 1.00 1.00 0.00 0.00 0.00
WHO noise (sustained) 1.00 1.00 1.00 1.00 1.00 0.00
Net community fiscal impact 0.15 0.00 1.00 0.00 0.00 0.00
Basin stress contribution 1.00 0.00 1.00 0.00 0.00 0.00
Generator runtime limit 0.50 0.00 0.50 1.00 0.00 0.00
WRI water stress 0.00 0.80 0.00 0.00 0.00 0.00
EJ screening 0.00 0.00 0.00 0.40 0.00 0.00
Transmission utilization 0.37 0.11 0.10 0.20 0.00 0.11
Ratepayer cost shift 0.00 0.00 0.00 0.00 0.32 0.00
N. Virginia
Tucson
C. Ohio
Memphis
Nordic
Saline